MORTON Table Salt (737g Canister)



Main Compound: Sodium Chloride (NaCl) AKA: Salt

Why is salt present?: Salt is used in cooking and baking as food flavoring/seasoning, preservation, etc., it is the main ingredient of this product along with some added ingredients

Molar-Mass Build-up:

Formula: NaCl | 1 Na + 1 Cl per formula unit

Element	Atoms	Atomic mass	Contribution
Na	1	22.99	22.99
Cl	2	35.45	35.45

Total molar mass = 58.44 g/mol

Mass composition: Na = 39.34%; Cl = 60.66%

Inside the Fomula:

For 1.50 g NaCl: $1.55x10^{22}$ formula units

Na atoms = $1.55x10^{22}$ x 1 = 1.55x1022 atoms Na

Cl atoms = $1.55x10^{22}$ x 1 = 1.55x1022 atoms Cl

Route 1: 1.50 g × 0.3934 = 0.590 g Na; 1.50 g × 0.6066 = 0.910 g Cl

Route 2: formula-unit count x each element's molar-mass share gives the same 39.34% Na and 60.66% Cl

Reverse formula check: 39.34/22.99 : 60.66/35.45 = 1 : 1 → NaCl

Sig-fig check and one-number takeaway:

Each 1.5 g Serving Contains about $1.55x10^{22}$ of NaCl

Base Quantitative Data:

Serving Size: ¼ Tablespoon (1.5 g)

Sodium: 590 mg per serving

Package size: 26 oz (737 g); about 491 servings

Ingredients: Salt, Calcium silicate

Atomic masses: Na = 22.99 g/mol; Cl = 35.45 g/mol

Model assumption: treat the 1.5 g serving as NaCl for the worked stoichiometry; actual morton salt also contains minor additives.

The Mole Map:

For 1.50 g NaCl:

Mass → moles: 1.50 g + 58.44 g/mol = 0.0257 mol

Moles → formula units: 0.0257 mol x $6.022x10^{23}$ = $1.55x10^{22}$ units

Reverse: formula units + 6.022x1028 = mol; mol x 58.44 g/mol = mass

Conversion factors: (1 mol NaCl / 58.44 g NaCl); (6.022×1023 formula units / 1 mol).

Scale-Up:

One full 26 oz package = 737 g (label-listed)

Moles/package: 737 g + 58.44 g/mol = 12.6 mol NaCl

Formula units/package: 12.6 mol x $6.022×10^{23}$ = $7.59x10^{23}$

Example (30 servings): 30 x 1.50 g = 45.0 g; 45.0 g + 58.44 = 0.770 mol

Packages per mole: 1 mol + 12.6 mol/package = 0.0793 package

References:

Disclosure: Used AI for drafting, research aid, and calculations aid. Final output is made in Canva.

Morton table salt | plain all-purpose salt for cooking & baking. (2026, April 27). Morton Salt. https://www.mortonsalt.com/shop/table-salt/morton-table-salt/